



Schooling and Development in Indonesia

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To what extent has Indonesia's expansion of schooling translated into measurable returns in wages and labor market participation, and why have gains been uneven across gender and region?

Background

Background on Indonesia

- A country in Southeast Asia and Oceania, between the Indian and Pacific oceans
- Population 283,487,931 (2024)
- GDP: 1.4 trillion with GDP growth around 5.0% in 2024
- Indonesia's poverty headcount ratio at \$3.00/day was 4.0% in 2025

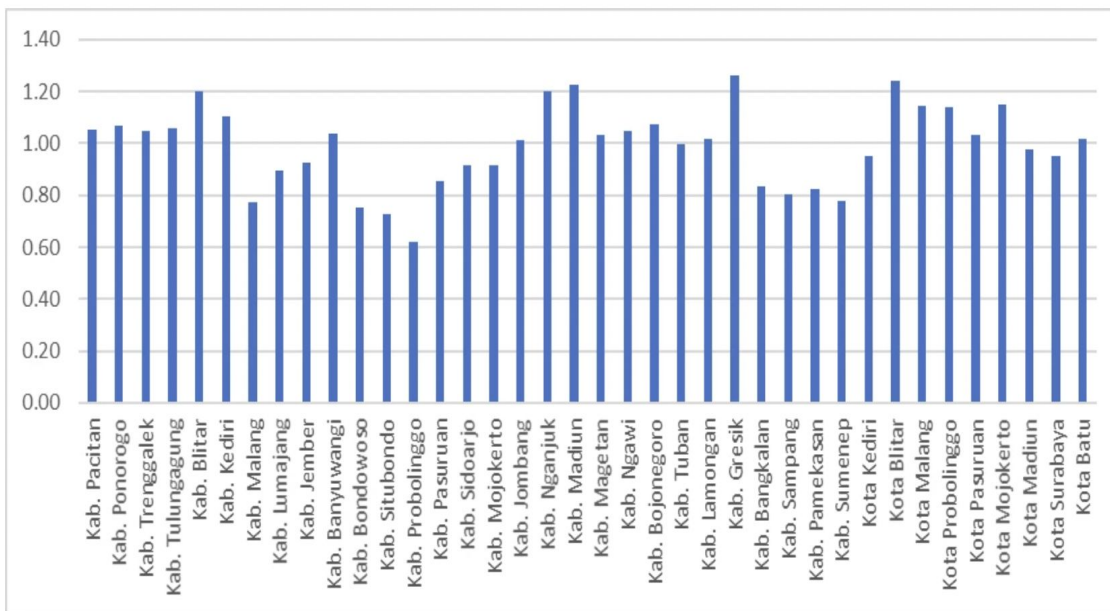


Background on Education

- In 1975, 65 percent of students were male
- GNI Index in 1970: 0.89 meaning girls were enrolled at lower rates compared to boys this can indicate that there's gender disparity
- GNI Index in 2019: 1.00, meaning there is gender parity with equal enrollment rates
- Since 2002, the primary and secondary school enrollment grew by more than 10 million students, or about 31%.
- Education gains also varied by region and the World Bank notes that subnational differences in gender parity still exist.



Gender Gaps in School Participation and Barriers to Education



Main reasons for dropping out from school by sex (percent)

Main Reason	Indonesia		
	Male	Female	Total
Insufficient funds	33.01	30.68	31.99
Working	18.74	13.68	16.52
Marriage	0.37	12.27	5.59
Taking care of the household	0.37	3.65	1.81
Assume that current education is enough	4.26	5.24	4.69
Shame of Poor Economic condition	2.88	1.9	2.45
School is far	3.71	3.26	3.51
Disability	5.39	4.82	5.14
Other	31.27	24.51	28.3
Total	100	100	100

Source: Susenas, 2017.

Policies and Reform

SD Inpres Program (1973~1978)

- The Catalyst: The 1973 Oil Crisis -> massive surge
- The Goal: President Suharto's administration aimed to achieve universal primary education by rapidly increasing school availability in rural and underserved areas
- Scale: Over 61,000 primary schools were built
- Key Features: **Top-Down Execution, Targeting the Poor, Standardization**





Subsequent Reforms: Strengthening the System (1990s – 2000s)

1994: 9-Year Compulsory Education Law

- Expansion: Extended mandatory schooling from 6 years (primary) to 9 years (junior secondary)
- Significance for Women: Targeted the critical transition period between primary and middle school

-> many girls dropped out due to social pressure, domestic expectation

2003: National Education Law (No. 20)

- The 20% Budget mandate: Requiring the government to allocate at least 20% of the national and regional budgets to education
- Institutionalization: Formalized education as a fundamental right for every citizen, providing a permanent financial shield for educational development regardless of political shifts.



Subsequent Reforms: Strengthening the System (1990s – 2000s)

2005: The BOS (Bantuan Operasional Sekolah) Program)

- Funding Model: A “School Operation Assistance” grant provided directly to schools based on enrollment numbers
- Targeting the Cost: Eliminating tuition fees for primary and junior high schools
 - > removed the primary obstacle for low-income families: **the cost of schooling**

Policy Design: Did Gender Equity Feature in the Design?

Gender-Neutral Design, Gender-Responsive Outcomes

- Most of Indonesia's education laws were drafted as gender-neutral; they applied to all citizens equally without specific "female-only" quotas

1. Targeting the "Marginalized" (Implicit Equity)

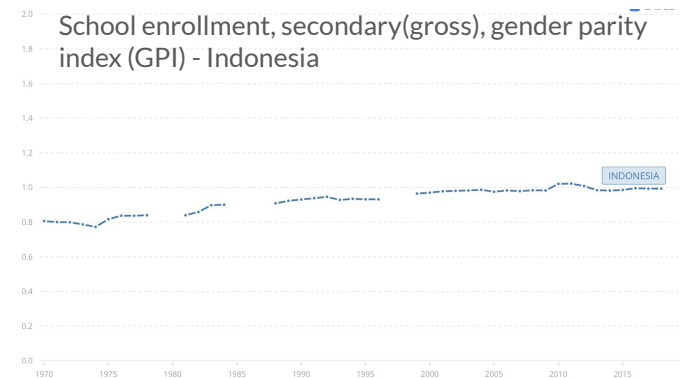
The reforms focused on **rural accessibility & low-income households**

Girls most likely to be kept home due to "Economic Son Preference" (families invest in limited resources in sons)

2. The result

By making education **free (BOS)** and **geographically accessible (Inpres)**

-> the government effectively neutralized the gender bias in family decision-making





Policies and Reform

- Indonesia successfully solve the 'Supply-side' problem
- Built the schools, passed the laws, and funded the classrooms
- Enrollment gaps have virtually disappeared

**Yet, did this massive investment in human capital pay off
for women's economic and social empowerment?**

Evidence of Returns

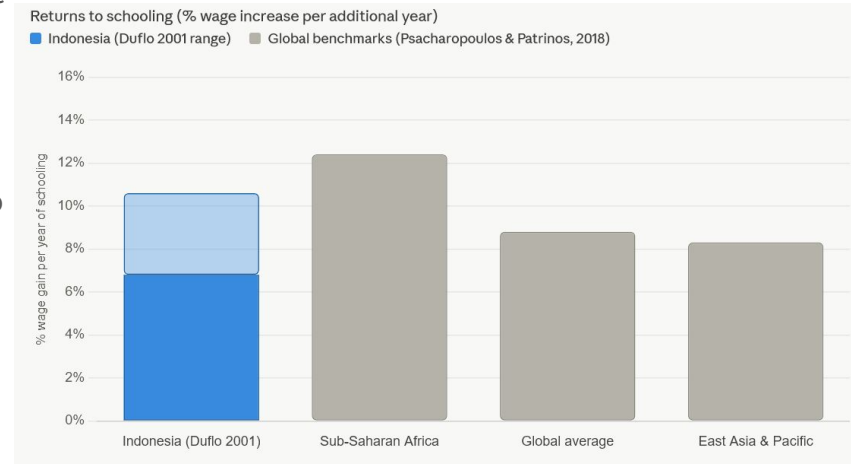


How We Measure Returns

- The core question: does one more year of school lead to higher wages, and by how much?
- The tool we can use is **mincer wage regression** which estimates the % wage increase per additional year of schooling
- **Problem:** People who get more education are different in the first place and it's hard to isolate the causation
- **Solution:** By Duflo, use INPRES as a natural experiment - regions that got more schools in 1973-78 are compared to regions that got fewer, across birth cohorts too old vs. young enough to benefit
- This works because school placement was based on prior enrollment rates, not wages, so it's as good as random for wage outcomes

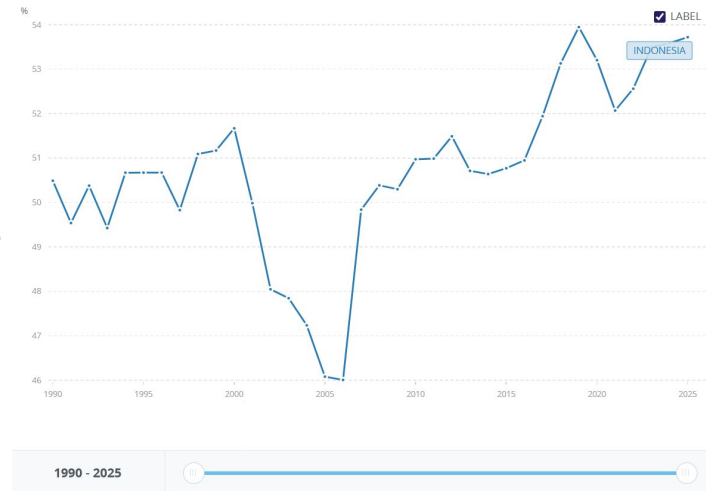
Aggregate Wage Returns

- **INPRES scale:** between 1973 and 1978, the government build over 61,000 primary schools, which roughly doubled Indonesia's stock
- Each additional school per 1,000 children increased years of education by 0.12 to 0.19 years per cohort exposed, and raised wages by 1.5% to 2.7%, according to Duflo
- The implied return to education was 6.8% to 10.6% wage increase per additional year of schooling, which is in line with global estimates
- **Bottom line:** INPRES causally increased both schooling and earnings, so the investment paid off when looking at the aggregate



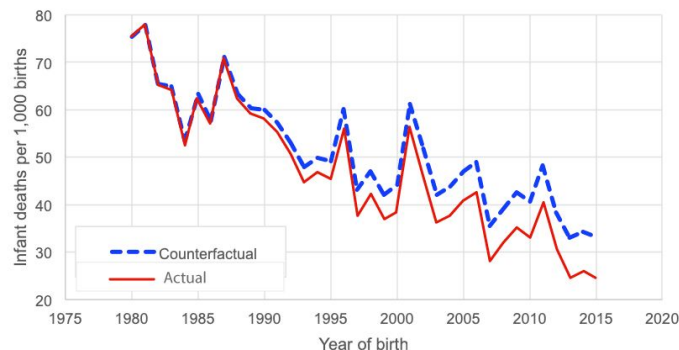
Gender-Disaggregated Returns

- **The paradox:** women's education levels have risen sharply yet the labor market returns haven't kept pace with them
- The labor force participation rate gap has barely moved in 20+ years, at around 54% for women and 85% for male, a 30 percentage point gap
- Women with the same level of education earn roughly 59 cents for every dollar a comparably education man earns, which is a 40% gap
- The reason for this is due to majority of women being in the informal sector in unregulated, lower-productivity, and lower-wage jobs, not the formal sector where schooling pays off the most
- The women in formal sector jobs often exit the labor force entirely after childbirth, suggesting formal employers fail to retain them



Beyond Wages: Health & Social Returns

Figure 1: Actual and counterfactual infant mortality rates by single year of birth among women in their first marriage at the time of the survey, assuming constant maternal educational distribution, Indonesia 1980–2015.



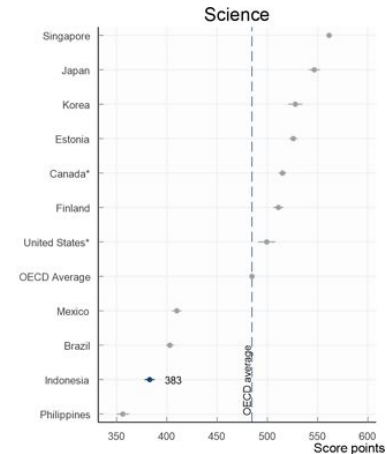
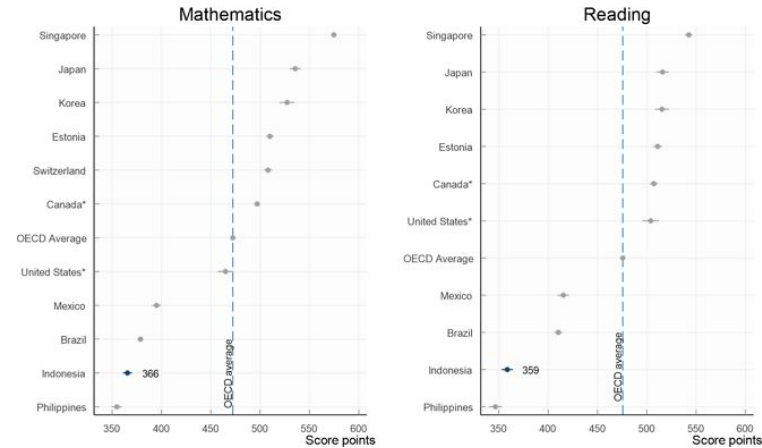
Source: Schellekens, J. (2021).

- **Key finding:** each additional year of maternal schooling reduces the probability of a child dying before age 1 by 0.6 percentage points, and before age 5 by 0.9 percentage points (ResearchGate, 2021)
- Reason for this is mothers make better healthcare decisions, have stronger voice in household resource allocation, and are more likely to seek prenatal care
- Female education is a stronger determinant of age at marriage and early fertility than male education, more schooling leads to later marriage, fewer children, and better child outcomes (OECD Dev Centre)
- Mother's schooling has a strong positive correlation with children's own schooling, and the returns compound across generations
- **The takeaway:** even where wage returns are unequal, health and social returns to female education are large and measurable, and these are undercounted in standard economic analyses

Remaining Challenges

More Schools, Not More Skills

- Indonesia built 61,000+ schools under INPRES (1973–1979), driving a dramatic rise in enrollment but construction outpaced curriculum and teacher investment
- Returns to schooling declined between 1993 and 2007, reflecting that credential supply grew faster than actual skill quality
- PISA 2022: Indonesia scored 366 in math and 359 in reading vs. OECD averages of 472 and 476, ranking 70th in math and 71st in reading out of 81 countries
- Indonesia consistently ranked among the lowest performers in Southeast Asia on international assessments (PISA, TIMSS)
- Teacher shortages remain severe: Jakarta suburbs reported 66% teacher deficits vs. 68% oversupply in city schools, same country, opposite problem





A Country of Two Education Systems

- Java and Greater Jakarta generate ~60% of Indonesia's GDP and absorb the majority of formal sector jobs, making returns to education geographically uneven even for equally-educated workers
- Decentralization post-1999 split education governance across 500+ districts, creating excellence in urban centers and near-collapse in remote ones
- Outer island regions (Papua, Maluku, East Nusa Tenggara) classified as "disadvantaged, frontier and outermost" (3T) face compounded barriers: no infrastructure, no teachers, no formal labor market nearby
- Education inequality is consistently worse in rural areas, and for rural women specifically
- Urban-rural wage gap exists even controlling for years of schooling: the same diploma yields lower returns outside Java

Educated but not employed

- Women now outperform men in higher education completion, yet only ~36.6% hold formal employment (2025)
- Female labor force participation: 56.7% in 2025, still trailing men by ~27 percentage points; was stagnant around 51% for nearly two decades prior
- 63% of working women are in the informal sector, no wage floors, no legal protection, no social insurance
- Gender wage gap: women earn ~23% less than men on average (2020 Labor Force Survey), driven by occupational sorting into lower-productivity roles
- Returns to schooling for women only jump sharply at the university level, junior and senior high school show weak gains in labor force participation





Structural Barriers

- Cultural norms frame women as household supporters, limiting entry into leadership and high-skill roles regardless of credentials; occupational segregation reflects social perception, not skill gaps
- Marriage and family expectations: participation rates are highest for single and divorced women, lowest for married women, suggesting domestic role expectations suppress formal employment
- Legal gaps compound the problem: no equal inheritance rights, weak workplace protections, and local regulations that conflict with national standards
- Education policy expanded access, but labor market policy never followed: no gender-targeted hiring incentives, weak enforcement of equal pay, and informal economy left largely unregulated
- World Bank (2020): many Indonesian labor programs carry "inherent bias," actively channeling women into lower-productivity jobs

Implications & Takeaways

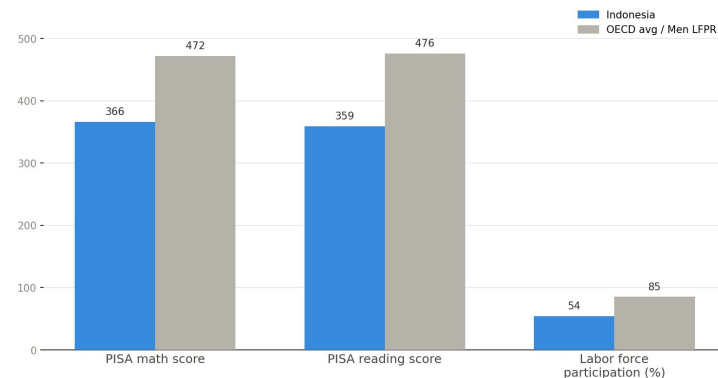


What the Evidence Says Works

- Targeted school construction (INPRES) raised wages 1.5-2.7% and years of schooling by 0.12-0.19 per cohort
 - Essentially infrastructure investment works when placed where gaps are the largest
- Compulsory education laws (1994 year mandate: 2003 National Education law) pushed secondary enrollment and reduced dropout rates
- BOS funding program reduced cost barriers for low-income families and helped close attendance gaps, especially for girls

What's Missing?

- Quality investment lagged quantity: PISA 2022 scores of 336(math) and 359(reading vs. OECD averages of 472 and 476, more schools didn't mean better learning
- Remote region enforcement: denteralization across 500+ districts created a near collapse in outer islands (papua, maluku and East Nusa Tenggara) despite the gains in Jakarta
- No gender- targeted labor policy: women gained schooling returns but still face occupational segregation, lower labor force participation and informal economy absorption
 - Education alone didn't fix structural barriers





The Takeaway

- Quantity of schooling is necessary but not sufficient
 - Indonesia proved you can build the schools and still fall short on wages, quality and equity
- Returns to schooling are real but uneven: strongest for men, urban workers and weakest for rural women and outer island regions
- Essentially:
 - Indonesia's school expansion translated into measurable wage returns
 - 6.8%-10.6% per additional year
 - But gains were uneven because policies targeted enrollment, not quality, labor markets, or gender barriers